

Eduardo Vasconcelos

Security Engineer, MSc in Computer Science, PhD candidate

Contact

- **E-mail:** eduardo@eduardovasconcelos.com
- **Website:** <https://eduardovasconcelos.com>
- **Codeberg:** <https://codeberg.org/vasconcedu>
- **GitHub:** <https://github.com/vasconcedu>
- **LinkedIn:** <https://www.linkedin.com/in/vasconcedu/>

Key Highlights

- Close to 10 years of experience in the software industry with emphasis on Security Engineering;
- MSc in Computer Science awarded for national-level excellence in Software Engineering research in 2024;
- Computer Engineer awarded for excellence in the undergraduate thesis in 2016;
- Experience in teaching programming in vocational/technical education;
- Proficient in three languages: Portuguese (*native*), English (*IELTS: C1*), and French (*TCF: B2-C1*).

Skills Matrix

Level	Application Security	Complementary
Advanced	Security Engineering, Threat Modeling, Penetration Testing, Vulnerability Assessment, Security Code Review	Research & Development
Intermediate	Reverse Engineering, Cryptography	Full Stack Development
Familiar	DevSecOps, Incident Response	Linux SysAdmin, Linux Kernel Development

Work History

- **Master Application Security Architect** @ SiDi, Brazil (remote)
Aug 2024–Present
 - Performed Android malware reverse engineering and developed malicious behavior detection heuristics and Machine Learning models;

- Research and development of automated Android malware analysis routines to unveil and monitor malicious behavior;
 - Conducted threat modeling, security architecture reviews, and assessments for Android and web applications, including Samsung Wallet and Samsung Knox;
 - Engaged in AI agent red teaming;
 - Collaborated closely with development teams to integrate security early into the Software Development Life Cycle (SDLC) and anticipate vulnerabilities;
 - Revitalized the company’s Security Champions program;
 - Designed and implemented security solutions, acting as an engineer for both backend and frontend applications;
 - Provided guidance and mentoring to less experienced peers.
- **Open Source Contributor @ Linux kernel**
Apr 2026–Present
 - Contributed 2 patches to the Linux kernel: 1 ThinkPad ACPI driver refactor (merged to maintainer’s next tree), 1 DRM subsystem code quality improvement (reviewed, awaiting merge);
 - Engaged with upstream review processes and kernel development workflows.
 - **Technical Instructor @ SENAI, Brazil (on-site)**
May 2025–Jan 2026
 - Taught Python programming and Microsoft Power BI courses.
 - **Senior Mobile Security Engineer @ iFood, Brazil (remote)**
Feb 2022–Aug 2024
 - Designed and implemented security and fraud prevention mechanisms for mission-critical Android and iOS applications with over 40 million active monthly users;
 - Reverse engineered malicious Android apps to reveal methods and behaviors used by fraudsters in food delivery services;
 - Integrated Runtime Application Self-Protection (RASP) solutions into GitLab CI/CD pipelines for mobile apps.
 - **Senior Software Security Analyst @ SiDi, Brazil**
Apr 2019–Feb 2022
 - Delivered ongoing offensive security testing services to Samsung, emphasizing application security for Android and web applications;
 - Conducted research on Android API and OS vulnerabilities on behalf of Samsung, including issues related to Android accessibility service abuses and permission circumvention.

- **Information Security Analyst @ Hacker Rangers, Brazil**

Jun 2017–Apr 2019

- Delivered comprehensive services in web application and network security assessments, as well as phishing campaigns;
- Developed, maintained, and administered Hacker Rangers, a SaaS cybersecurity awareness platform;
- Installed, configured, provided training for, and managed IBM Security Guardium database audit and protection virtual appliances, supporting both private organizations and government agencies.

- **Information Technology Intern @ Embraer, Brazil**

Sep 2016–Jun 2017

- Contributed to the development of an IT Change Management plan;
- Conducted analyses of corporate web traffic.

- **Engineering Intern @ SigMEDIA, Ireland**

May 2014–Jul 2014

- Developed a machine learning model designed to detect laughter in human speech;
- Processed digital voice signals to train the model.

Education

- **Doctoral Degree in Computer Science @ USP, Brazil**

Apr 2026–Present

- Research topic: fuzz testing embedded systems operating in brownout voltage regimes for security bugs.

- **Master’s Degree in Computer Science @ USP, Brazil**

Feb 2022–Aug 2025

- Graduated with an A average;
- Research topic: leveraging mutation testing techniques to inject vulnerabilities into Android apps, providing a novel approach to evaluating mobile security testing procedures;
- Received the Distinguished Paper Award at the Brazilian Conference on Software 2024 (CBSOFT’24) for my work, “Mutation Testing to Support the Security Testing of Android Applications”;
- Served as a teaching assistant for Software Engineering.

- **Postgraduate Diploma in Software Engineering @ Unicamp, Brazil**

Feb 2019–Dec 2019

- Graduated with a GPA of 9.4 out of 10.0;

- One-year Postgraduate Diploma program in Software Engineering, covering a range of disciplines within the field.

- **Engineer’s Degree in Computer Engineering @ USP, Brazil**

Feb 2012–Jul 2017

- Graduated 13th out of 59 with a GPA of 7.5 out of 10.0;
- Awarded the Sao Paulo Research Foundation (FAPESP) undergraduate research scholarship, contributing to a research project focused on flood forecasting in urban areas;
- Received the Best Computer Engineering Undergraduate Thesis Award in 2016 for my work on emotion recognition in human speech using Machine Learning;
- Served as a teaching assistant for Python Programming;
- Served as a teaching assistant for Object-Oriented Programming;
- Volunteered as a junior enterprise web developer.

- **Visiting Student in Electronic Engineering @ TCD, Ireland**

Sep 2013–May 2014

- Served as a full-term visiting student, earning 60 ECTS with honors;
- Awarded the Brazilian Federal Government’s “Science without Borders” (SwB) scholarship;
- Received a Special Prize for the Best Product Branding Video at the Trinity Engineering Student Entrepreneurs of the Year Competition in 2014.

- **Professional Training in Computer Hardware and Network Maintenance @ Microlins, Brazil**

Mar 2007–Sep 2007

- Six-month professional training program in computer hardware, operating systems, and computer networks.